

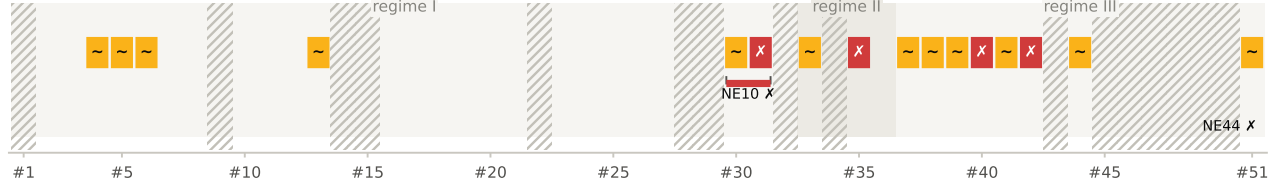
H30: An operator-susceptibility gauge χ_{op} : how steerable is the swarm today?

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 2/5 Usefulness 2.5/5

Bottom line. Operator messages have a period-level susceptibility (about one active minute per nudge; a small content pull), but daily steerability is not a trackable state.

level: hypothesis · pipeline checks 86% · updated 2026-10-04

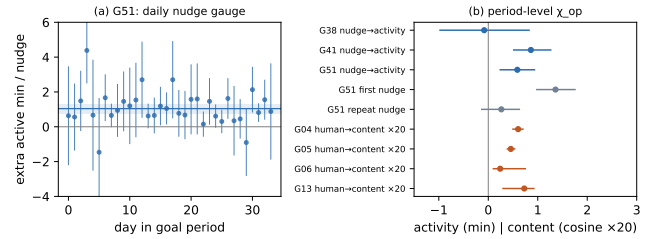


Goal periods tested (6 failed, 12 mixed). ✓ supported ✗ failed ~ mixed d descriptive ... pending.

Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

We measure how far one operator message moves the agents who see it, and whether that number is trackable by day. We test 16 non-holdout goal periods, after a synthetic validation that fixed three estimator biases. **Round 1b** uses the corrected activity table, because round 1 lost half of all events. It adds the leading-@ nudge target, receiving-call timing and past-only controls with day effects. *Activity*: a nudge adds about one active minute for its target (#51 0.98 [0.63, 1.37], #38 0.95 [0.38, 1.54]; round 1: 0.59, -0.08). First nudges give 1.21, repeats 0.44, room-mates ≈ 0 . The round-1 lull warning reverses (lull 1.74 vs 0.72). *Content*: human messages turn the next statements toward their new content, +0.024 (room-mates) and +0.041 (named), in both embedding models. *Daily gauge*: #51 varies by day ($p = 0.026$, $R_1 = 0.48$) without persistence. NE44: nudges buy more under the 5-min pause default (reversed). The holdout is not run.



Observables. (a) #51 daily nudge gauge: extra active minutes per nudge for the target (day FE, 95% CIs), with the period mean and band. Day-to-day scatter matches the daily CIs: no trend, no persistence. (b) Period-level χ_{op} . Nudge→activity in the three largest nudge periods (pre-registered model); first vs repeat nudges in #51 (day FE); human→content in human-dense periods (cosine $\times 20$).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not watch a daily “steerability” number: at village message rates it is a period constant plus noise, and it does not age. One nudge to an idle agent buys about one extra active minute from its receiving call. A first nudge buys 1.2 min and a repeat 0.4 (CI includes 0). Nudging into a swarm-wide lull works even better (1.74 vs 0.72 min), and short scaffold pauses help (NE44). To change *what* an agent writes, name it and add new content (+0.041 vs +0.024 for room-mates). For analysts: do not condition on future nudges, as H04’s isolation rule did.

PHYSICAL MODEL

Linear response to impulsive fields (01, 11). Each operator message k is a field pulse on its recipients i . Spins: activity $n_i(t) \in \{0, 1\}$; content s_i , a whitened statement embedding. $u_k^\perp$: the message direction projected off the recipient’s last 8 statements. χ_{act} : the 30-min integrated activity response (local projection; strata α_s , day effects δ_d , kick counts K). χ_{con} : post-message alignment minus same-kind placebos p . $\chi_{\text{op}}(d)$: the day mean of per-kick responses r_k .

$$Y_i(m) \equiv \sum_{\tau=1}^{30} n_i(m + \tau) = \alpha_{s(i,m)} + \delta_d + \sum_c \chi_{\text{act}}^c K_i^c(m) + \beta \cdot K_i^{\text{adj}}(m) + \varepsilon$$

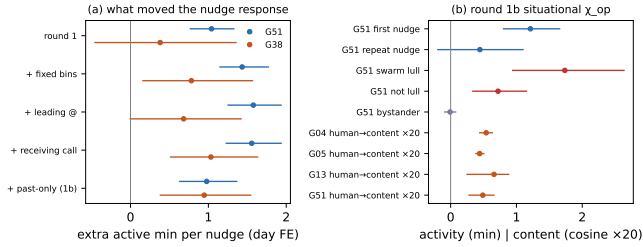
$$\chi_{\text{con},k} = \bar{s}_i^{\text{post}} \cdot u_k^\perp - \langle \bar{s}_i^{\text{post}} \cdot u_p^\perp \rangle_p, \quad u^\perp = \frac{(1 - P_{\text{pre}})u}{\|(1 - P_{\text{pre}})u\|}$$

$$\chi_{\text{op}}(d) = \frac{\sum_{k \in d} w_k r_k}{\sum_{k \in d} w_k}, \quad R_1 = 1 - \frac{\text{Var}_{\text{perm}}[\chi_{\text{op}}(d)]}{\text{Var}[\chi_{\text{op}}(d)]}$$

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RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b	Verdict
P1 nudge→target	#51 0.59 → 0.98 [0.63, 1.37]; #38 -0.08 → 0.95 [0.38, 1.54]; small periods 5/10 > 0	✓ (count rule ×)
P2 bystanders ≈ 0	#51 0.13 → -0.01; #38 -0.54 → -0.09	✓
P3 human→activity	≤ 0 / n.s. in regime I (unchanged)	×
P4 human→content	0.026 → 0.024 (gte 0.024); named 0.046 → 0.041	✓
P5 nudge→content	#51 0.003 → 0.002 [-0.001, 0.005]	✓
P6 no daily state	#51 p 0.08 → 0.026, R_1 0.28 → 0.48, lag-1 0.08	× as written
P7 no aging	#51 +0.02 [-0.03, 0.07]/day	✓
P8 context saw-tooth	pull falls 0.66 vs late 1.49	×
P10 family	no Holm-significant lab	✓
P11 nulls	#51 and #38 now pass	✓
Native #5 dose	#51 +0.02 → 0.013 over 1–7 msgsg; 8+ noisy	× (rule)
Native NE44	12-h 0.57 vs 5-min 0.98 (Δ -0.41)	× reversed
Native NE10	first nudges ever -0.42 (22 kicks)	× low power



Second figure. Round 1b. (a) What moved the nudge response in #51 (blue) and #38 (orange). Restoring the dropped events carries most of the change; past-only controls bring the level back to ≈ 1 min. (b) Situational χ_{op} in #51 (first vs repeat nudges, swarm lull vs not, bystanders) and human content pull ($\times 20$).

GOAL PERIODS IN DETAIL

G04 ~ 1b: mixed (r1 mixed; fixed bins, receiving call)
 G05 ~ 1b: mixed (r1 mixed) · native dilution: failed
 G06 ~ 1b: mixed (r1 mixed; fixed bins, receiving call)
 G13 ~ 1b: mixed (r1 mixed; fixed bins, receiving call)
 G30 ~ 1b: mixed (r1 mixed; fixed bins, receiving call)
 G31 × 1b: failed (r1 mixed; fixed bins, receiving call)
 G33 ~ 1b: mixed (r1 mixed; fixed bins, receiving call)
 G35 × 1b: failed (r1 failed; fixed bins, receiving call)
 G37 ~ 1b: mixed (r1 mixed; fixed bins, receiving call)
 G38 ~ 1b: mixed (r1 mixed; fixed bins, receiving call)
 +8 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Only #51 and #38 resolve the nudge response; 3–5-day periods give unstable CIs. The level depends on the future-kick adjustment: 0.98 past-only vs 1.56 with the A1 regressors of round 1 (day FE). SE-based window reliabilities exceed 0.5 in #51 but also in #38, where the calibrated permutation finds no day signal, so we do not count them. Human effects on activity in dense regime-I chat are not identified. Content measures message-specific steering only, and human replies may share topics. The round-1b design (receiving-call timing, past-only, day FE primary) follows the re-evaluation brief, not the pre-registration.

ROUND 2

- **Essence.** a trustworthy answer to "will my next message change what this swarm does?". Round 1 says it is per situation, not a daily state.
- **H30-R1** Situational χ instead of daily χ .
- **H30-R2** The 2026-08-20 switch-off as a natural experiment.
- **H30-R3** Embedding swap and reply control for χ_{con} .
- **H30-R4** Per-message content strength.
- **H30-R5** Run confirm.py after the card is committed.

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